

Tutorial 4

Elementary matrices, singularity and invertibility

1. Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 5 & 4 & 5 \\ 2 & 0 & 3 \end{pmatrix}.$$

- (a) Use the algorithm in the notes for Gaussian elimination to get A into row echelon form.

Solution: We need two row operations: Subtract 5 times row 1 from row 2 and subtract 2 times row 1 from row 3. We obtain matrix $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, which is in row echelon form.

- (b) Is A invertible? Use your answer to (a) only to give a reason for your answer. (Do not try to find the inverse!)

Solution: A is invertible because each column in the row echelon form obtained in (a) has a pivot.

- (c) Now find the **reduced** row-echelon form of A .

[Hint: Use the row-echelon form in (a) and then carry on in a suitable way to make all the pivots equal to 1 and get 0's above the diagonal as well, if possible.]

Solution: We take matrix $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ from (a), multiply row 2 by $1/4$ and subtract row 3 from row 1. We obtain the matrix, $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, which is the reduced row-echelon form of A .
(We thus obtained the identity matrix I_3 as the reduced row echelon form. Recall that unlike row echelon form, the **reduced** row echelon form is always **unique**.)
(As you can see, the order of these two row operations does not matter in this case.)

- (d) For every elementary row operation e you have used, write down the corresponding elementary matrix $E = e(I_3)$.

Solution:

Let E_1 be the elementary matrix corresponding to the first elementary operation in (a), E_2 be the elementary matrix corresponding to the second elementary operation in (a), E_3 be the elementary matrix corresponding to the first elementary operation in (c), E_4 be the elementary matrix corresponding to the second elementary operation in (c).

$$\text{Then } E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix},$$

$$E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_4 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Note: If you are asked a question like this, there is no need to multiply out $E_4 E_3 E_2 E_1 A$ to show that you get I_3 ; you will get no extra marks for it! If you have plenty of time and want to check your calculations, you may of course do this, but do not waste time on it in a test.

- (e) Use your answer to (d) to find elementary matrices E_1, E_2, E_3 and E_4 such that $E_4 E_3 E_2 E_1 A = I_3$. Make sure you get the E 's in the correct order.

Solution: Take above matrices E_1, E_2, E_3 , and E_4 .

- (f) Write A as a product of elementary matrices. [Hint: Look at (e).]

Solution: It follows from $E_4 E_3 E_2 E_1 A = I_3$ that

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} \quad (\text{note the order})$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

To find the inverses of the elementary matrices above, we use the fact that if $E = e(I_n)$ is an elementary matrix, then $E^{-1} = e^{-1}(I_n)$, where e^{-1} is the inverse of the row operation e .

- (g) Write A^{-1} as a product of elementary matrices.

[*Hint:* How do we get the inverse of a product of invertible matrices?]

Solution: $A^{-1} = E_4 E_3 E_2 E_1$, using the matrices in (d) above and Theorem 1.5.7.

2. Let $A = \begin{pmatrix} 0 & 4 & 1 \\ 2 & 7 & 0 \\ -4 & -6 & 2 \end{pmatrix}$.

- (a) Find elementary matrices E_1, E_2 and E_3 and a matrix U in row-echelon form such that $E_3 E_2 E_1 A = U$.

Solution: Do Gaussian elimination on A to obtain its row echelon form:

$$\begin{aligned} \begin{pmatrix} 0 & 4 & 1 \\ 2 & 7 & 0 \\ -4 & -6 & 2 \end{pmatrix} &\sim \begin{pmatrix} 2 & 7 & 0 \\ 0 & 4 & 1 \\ -4 & -6 & 2 \end{pmatrix} && R_1 \leftrightarrow R_2 \\ &\sim \begin{pmatrix} 2 & 7 & 0 \\ 0 & 4 & 1 \\ 0 & 8 & 2 \end{pmatrix} && R_3 \rightarrow R_3 + 2R_1 \\ &\sim \begin{pmatrix} 2 & 7 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 0 \end{pmatrix} && R_3 \rightarrow R_3 - 2R_2 \end{aligned}$$

Therefore we can take

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$$

and $U = \begin{pmatrix} 2 & 7 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

- (b) Is the matrix U you obtained non-singular? Give a reason for your answer.

Solution: No. There is no pivot in the third column of U . It follows from this that the system of equations $U\mathbf{x} = \mathbf{0}$ has a free variable, and therefore has infinitely many solutions. Hence U is singular.

- (c) Can A be written as a product of elementary matrices? Give a reason for your answer.

Solution: No. The system $A\mathbf{x} = \mathbf{0}$ has the same solution set as the system $U\mathbf{x} = \mathbf{0}$, since A and U are row equivalent. From (b) there are infinitely many solutions, so A is singular, by definition, and it can therefore not be written as a product of elementary matrices, by the Invertibility Theorem.

3. Let $A = \begin{pmatrix} 1 & 0 & -2 \\ -3 & 1 & 2 \\ 2 & -3 & 4 \end{pmatrix}$.

- (a) Use the algorithm introduced in class to determine whether A is invertible, and if it is, find A^{-1} .

Solution: We know that if A^{-1} exists, then $[A, I]$ can be row reduced to $[I, A^{-1}]$, while if A is not invertible then A cannot be row reduced to I .

$$\begin{aligned} [A, I] &= \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 2 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array} \right) \\ &\sim \text{Row reduce (show your work)} \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -5/2 & -3/2 & -1/2 \\ 0 & 1 & 0 & -4 & -2 & -1 \\ 0 & 0 & 1 & -7/4 & -3/4 & -1/4 \end{array} \right). \end{aligned}$$

As a consequence, we know that

$$A^{-1} = \begin{pmatrix} -5/2 & -3/2 & -1/2 \\ -4 & -2 & -1 \\ -7/4 & -3/4 & -1/4 \end{pmatrix}.$$

Now answer the following two questions without doing any further Gaussian elimination, using only your answer to (a).

- (b) Find the solution set for the system of equations $A\mathbf{x} = \mathbf{0}$.

Solution: Because A is invertible, the only solution is $\mathbf{0}$.

- (c) Find the solution set for the system of equations $A\mathbf{x} = \mathbf{b}$ where $\mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix}$.

Solution: $\mathbf{x} = A^{-1}\mathbf{b} = \begin{pmatrix} -5/2 & -3/2 & -1/2 \\ -4 & -2 & -1 \\ -7/4 & -3/4 & -1/4 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -9 \\ -14 \\ -13/2 \end{pmatrix}.$

4. Let A be an $m \times n$ matrix, and suppose it is row reduced to a matrix U in row echelon form. Note that A is **not** necessarily a square matrix, and that U is a row-echelon form of A , not of an augmented matrix. Are the following statements true or false? If true, give a proof. If false, give a counter-example.

[**Warning:** The Invertibility Theorem is a theorem about **square** matrices.]

- (a) If there is a pivot in every column of U , then the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$.

Solution: False. For example, we may have

$$U = \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{4} & 5 \\ 0 & 0 & \boxed{6} \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 7 \\ 8 \\ 9 \\ 10 \end{pmatrix}.$$

The corresponding system of linear equations is represented by the augmented

matrix $\left(\begin{array}{ccc|c} \boxed{1} & 2 & 3 & 7 \\ 0 & \boxed{4} & 5 & 8 \\ 0 & 0 & \boxed{6} & 9 \\ 0 & 0 & 0 & 10 \end{array} \right)$ and has no solution (because of the last row).

- (b) If the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$, then the solution (for each $\mathbf{b}$) is unique.

Solution: False. For example, the system $\left(\begin{array}{ccc|c} \boxed{1} & 2 & 3 & b_1 \\ 0 & \boxed{4} & 5 & b_2 \end{array} \right)$ has infinitely many solutions for every $b_1, b_2 \in \mathbb{R}$.

- (c) If the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for *some* $\mathbf{b}$, then it has a unique solution for *every* $\mathbf{b}$.

Solution: False. For example, the system $\left(\begin{array}{ccc|c} \boxed{1} & 2 & 3 & b_1 \\ 0 & \boxed{4} & 5 & b_2 \\ 0 & 0 & \boxed{6} & b_3 \\ 0 & 0 & 0 & b_4 \end{array} \right)$ has a unique solution when $b_1 = b_2 = b_3 = b_4 = 0$, but no solution when $b_1 = b_2 = b_3 = b_4 = 1$.

- (d) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for *every* $\mathbf{b}$ if and only if U has no zero rows.

Solution: True. If U has no zero rows then the system is consistent for every $\mathbf{b}$ (we showed this using back-substitution). For the converse, suppose

the augmented matrix $[A|\mathbf{b}]$ is row reduced to $[U|\mathbf{b}']$. Then $U = E_k \cdots E_1 A$ and $\mathbf{b}' = E_k \cdots E_1 \mathbf{b}$ for some elementary matrices $E_1, \dots, E_k$. If the last row of U is zero, then choosing any $\mathbf{b}'$ whose last entry is non-zero gives a vector $\mathbf{b} = E_1^{-1} \cdots E_k^{-1} \mathbf{b}'$ for which the original system has no solution. Since this contradicts our assumption, U must have no zero rows.

- (e) Now answer each question again under the assumption that A is **square**.

Solution: Now all true! This is because if U is square then the number of pivot columns is equal to the number of non-zero rows. So U has no zero rows if and only if there is a pivot in every column of U .

5. Use the **definitions** and the **theorems** about invertible matrices in section 1.5 and 1.6 to prove the following:

- (a) If B is invertible, then B^{-1} is invertible.

[*Hint:* Your proof need not be longer than one line. Find the inverse of B^{-1} ; look at the definition of invertibility to see what you need for B^{-1} to be invertible.]

Solution: Since B is invertible, $B^{-1}B = I = BB^{-1}$. Therefore there is a $C(= B)$ such that $B^{-1}C = I = CB^{-1}$. By definition, B^{-1} is invertible.

- (b) If AB is non-singular, then B is non-singular.

[*Hint:* Use the definition of a non-singular matrix. You may find it easier to prove the contra-positive of this statement.]

Solution: Suppose $B\mathbf{x} = \mathbf{0}$. Then $A(B\mathbf{x}) = A\mathbf{0} = \mathbf{0}$, so $(AB)\mathbf{x} = \mathbf{0}$. But, since AB is non-singular, this means that $\mathbf{x} = \mathbf{0}$. Since we have shown that $B\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$, B is non-singular.

- (c) If A and B are square matrices and AB is invertible, then both A and B are invertible. [*Hint:* Use your answers to (a) and (b).]

Solution: Important note: Theorem 1.5.7 says that **if** A and B are invertible, **then** AB is invertible and in this case $(AB)^{-1} = B^{-1}A^{-1}$. If we only know that AB is invertible, we **cannot** conclude that its inverse is $B^{-1}A^{-1}$ before we know that both A and B are invertible. Put in another way, **we do not know that the converse of the statement in Theorem 1.5.7 holds**. Once we have proved the statement in (c), we will know that the converse holds, but we cannot use the converse before we have proved it!

Here is a proof of (c): Since AB is invertible, AB is non-singular, and therefore by (b) B is non-singular, and hence invertible. Then B^{-1} is invertible by (a), and so $A = A(BB^{-1}) = (AB)B^{-1}$ is invertible by Theorem 1.5.7.

- (d) If A and B are square matrices such that $AB = I$, then $BA = I$. (From this it follows that in the definition of the inverse of a square matrix A , it is enough to require that there be a square matrix B such that $AB = I$.)

[Hint: Use (c) and the fact that I is invertible.]

Solution: Since $I = AB$ is invertible, it follows from (c) that A is invertible. Hence $AB = I \Rightarrow A^{-1}AB = A^{-1}I \Rightarrow B = A^{-1} \Rightarrow BA = I$.

- (e) Now write down the converse of the statement in 5(b) and prove or disprove it.

Solution: If B is non-singular, then AB is non-singular. This statement is false. For a counter example, take $B = I_2$ and A any singular 2×2 matrix.

6. This question gives you an idea of how we'll test your understanding of proofs.

Read the following proof and then answer the questions following it.

Proof. Let A be an $n \times n$ non-singular matrix. We show that A is row equivalent to I_n . We know that A is row equivalent to a matrix U in row echelon form. (1)

The matrix U must look like this:

$$\begin{pmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ 0 & 0 & \dots & \\ \vdots & & \dots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{pmatrix}.$$

If we can show that the u_{ii} are non-zero for $1 \leq i \leq n$ then we can carry on row reducing until we obtain I_n . (2)

Suppose that one of the u_{ii} 's is zero. Then, because U is a square matrix in row echelon form, u_{nn} must be zero too. (3)

But that means that we have at most $n - 1$ pivots, so at least one free variable in the equation $U\mathbf{x} = \mathbf{0}$. (4)

It follows from this that the equation has infinitely many solutions. (5)

But since U and A are row equivalent, we have by a previous result that $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions. (6)

This is a contradiction of our assumption, so we must have $u_{ii} \neq 0$ for every i . (7)

Since we now know that U can be row-reduced to I_n , the result is proved. $\square$

- (a) Carefully state the result proved here.

Solution: Every non-singular $n \times n$ matrix is row equivalent to I_n .

- (b) Why can we make the statement at (1)?

Solution: This follows from the fact that every matrix can be row-reduced to a matrix in row-echelon form, and is hence row equivalent to such a matrix.

- (c) Explain why it is possible to perform elementary row operations on U which will make all the entries above u_{ii} zero provided $u_{ii} \neq 0$. How does it follow from this that we can row-reduce U to I_n ?

Solution: For $k < i$ we can make u_{ki} zero by subtracting u_{ki}/u_{ii} times row i from row k . If each $u_{ii} \neq 0$, then every entry above the main diagonal can be made zero. Then the i -th diagonal entry can be made 1 by multiplying row i by u_{ii}^{-1} .

- (d) Why can we say, at (3), that $u_{nn} = 0$?

Solution: If $u_{ii} = 0$, the pivot in row i must be in a column to the right of column i . Since each pivot is to the right of the pivot above it, there cannot be a pivot in row n , i.e. every entry in row n is zero.

- (e) Why can we say, at (4), that there are at most $n - 1$ pivots?

Solution: Row n cannot have a pivot.

- (f) Explain why the fact that there is a free variable in the equation $U\mathbf{x} = \mathbf{0}$ means that the equation will have infinitely many solutions.

Solution: If there is a free variable in the equation $UX = \mathbf{0}$, then we can solve for the dependent variables in terms of the free variable. For each real number t , we then get a solution by putting the free variable equal to t ; in this way we obtain infinitely many solutions, one for each value of t .

- (g) State the “previous result” used at (6).

Solution: We have shown (Lemma 1.6.10 in the notes) that if A and B are equivalent matrices, then $AX = \mathbf{0}$ and $BX = \mathbf{0}$ have exactly the same solutions. Hence if A and U are row equivalent and $U\mathbf{x} = \mathbf{0}$ has infinitely many solutions, then $A\mathbf{x} = \mathbf{0}$ must have infinitely many solutions.

- (h) What is the contradiction referred to at (7)?

Solution: The fact that $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions contradicts the assumption that A is non-singular.

- (i) Explain why the result is proved once we know U can be row-reduced to I_n .

Solution: We now know U is row equivalent to I_n . Since A is row-equivalent to U , it then follows that A is row-equivalent to I_n .